

1871 Airy: Supposed Alteration in Astronomical Aberration by a Refracting Medium — verification companion

George Biddell Airy

Airy 1871 — verification companion

[!abstract] Legend - □ tip/note — what the source establishes and how it is used - □ quote — verbatim source text

[!success] PDF present `project_sources/1871_Airy_Supposed_Alteration_Aberration.pdf` is in this folder — check the claims below directly.

Bibliographic Info

- **Author:** George Biddell Airy
- **Title:** On a supposed alteration in the amount of astronomical aberration of light, produced by the passage of the light through a considerable thickness of refracting medium
- **Year:** 1871
- **Publication:** Proceedings of the Royal Society of London, vol. 20, pp. 35–39
- **DOI:** 10.1098/rspl.1871.0011
- **Archive:** archive.org/details/paper-doi-10_1098_rspl_1871_0011 ; royalsocietypublishing.org/doi/10.1098/rspl.1871.0011

Claim used in the sheet / notebook

[!note] The one number taken from Airy The aberration constant is **unchanged** when the telescope is filled with water: the same $\approx 20.5''$ is read in air and in water. This is the null result Airy set out to test (prompted by Klinkerfues' claim of a change). It is the measured (Airy), air & water = $20.5''$ cell and the benchmark the models are scored against.

[!tip] How it feeds the argument Airy expected the corpuscular/light-slower prediction $n\beta \approx 27''$ if Earth's motion carried the light through a moving medium. He measured no change from the air value. In the sheet: the WATER block shows $27''$ (classical, no drag) and $20.5''$ (measured) side by side — the gap the heliocentric reading must close with Fresnel drag + the x_n calibration.

[!warning] Verify against the PDF Confirm from the paper text: (1) the water column length actually used, (2) that the measured constant matches the air value to within his stated error, (3) his own statement of the expected change he was testing for. Pull the exact sentence stating the null before quoting it anywhere published.

Backlinks

- [[2025_Ligabue_Airy_Relativity]]
- [[1867_Klinkerfues_Aberration_Wave_Theory]]

Tags

#source #verify